

LEONARDO PÉREZ

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EDUCATION

Tecnológico de Monterrey

Ciudad de México, México

Bachelor of Science, Computer Science

Aug 2023 – Jul 2027

Relevant Courses: Software development and decision making, Data Structures and Algorithms, Object-Oriented Programming, Security integration in networks and systems.

EXPERIENCE

Beholder

Syntax Challenge, HackPuebla 2025

Puebla, México

Aug 2025

- Won the Syntax Challenge at HackPuebla 2025 by developing an AI-powered parental control system.
- Built a Python client application with a keylogger to capture and filter chat interactions in real time.
- Deployed a FastAPI backend on Google Cloud using WebSockets and HTTP requests to process and store incidents.
- Integrated MongoDB Atlas for secure storage of flagged interactions, applying hashing for privacy.
- Designed and developed a React dashboard for parents to monitor live activity, flagged conversations, and incident severity.

Identifeye

Liverpool challenge, HackMX 2024

Estado de México, México

Oct 2024

- Won the liverpool challenge at HackMX 2024 by creating a scalable product recommendation system.
- Developed a full-stack web platform with React, Firebase and Flask, deployed as microservices on Google Cloud run.
- Implemented vector embeddings for 70,000 products, enabling similarity-based search with sub-second latency.
- Reduced runtime by 40% through optimized use of parallelism and concurrency in backend microservices.

Evergreen

Guadalalahacks Hackathon

Jalisco, México

May 2024

- Achieved first place at Guadalalahacks 2024 by developing a web application that identifies optimal tree-planting areas using geographic imagery.
- Improved terrain analysis by integrating YOLO with OpenCV to classify land types from uploaded images.
- Enabled accurate geolocation by using Geopy and Google Maps API to extract exact coordinates and terrain elevation.
- Facilitated reforestation planting by automating calculations with trigonometric and algebraic models, reducing manual input by 90%.

PROJECTS

Operación Escape

- Created a hybrid educational system for a local school combining a Unity game and React dashboard to enhance student engagement.
- Improved user experience by designing intuitive interfaces tailored for children and school staff.
- Streamlined deployment using Docker and Docker Compose to deliver a portable, installation-free solution.

Text-based Dungeons&Dragons Game

- Engineered a full game system from scratch in C++ using advanced templates and object-oriented design.
- Built custom libraries to manage catalogs, inventories, and maps with no external dependencies.
- Promoted code reuse and modularity via clean architecture and design patterns.

SKILLS

Programming Languages: Python, C++, Javascript, Go, Swift, Bash

Frameworks: React, FastAPI, Nextjs, Nestjs, SwiftUI, Gin

Tools: Docker, Git, AWS(EC2, S3), Linux, MySQL, MongoDB

Languages: English(C1), Spanish(Native), French(A2)